# 📘 1-Year Roadmap: Data Analytics, Data Science & GenAI

### 🎯 Goal: Placement-Ready for roles like Data Analyst, Data Scientist, GenAI Engineer, LLM App Developer

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## ✅ Module 0: Orientation & Core Concepts

🕒 Duration: 1–2 Days

🎯 Goal: Get students familiar with industry terms, career paths, tools, and how everything connects.

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### 🔷 Basic Industry Terms to Understand

- What is **Data** (Structured, Semi-Structured, Unstructured)
- Difference between **Data Analytics**, **Data Science**, **Machine Learning**, and **Artificial Intelligence**
- What is **Generative AI** (ChatGPT, DALL·E, etc.)
- What is **Cloud Computing** and why it matters in Data/AI
- What are **Databases** (SQL vs NoSQL)

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### 🔷 Career Roles You Can Target

- Data Analyst
- BI Analyst
- Data Scientist
- ML Engineer
- Gen AI Engineer
- Prompt Engineer
- Data Engineer

- Cloud Data Engineer
- AI Product Analyst

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## 🔷 Tools You'll Learn Throughout the Course

| Category | Tools You'll Use |
|---|---|
| Programming | Python, SQL |
| Analysis & BI | Excel, Power BI |
| Data Science & ML | Pandas, NumPy, Scikit-learn, XGBoost |
| GenAI & Prompting | ChatGPT, OpenAI API, LangChain, Hugging face, Streamlit, Agno |
| Databases | MySQL, PostgreSQL |
| Cloud & Deployment | AWS (S3, EC2, RDS), Azure basics, bedrock, Docker |
| Version Control | Git & GitHub |

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# ✅ Module 1: Python Programming for Data & AI (4 weeks)

**Duration:** Month 1

**Topics:**

- Python basics: variables, loops, conditions, functions
- Data structures: list, dict, set, tuple
- File handling, JSON, APIs
- OOPs basics
- Exception handling

**Tools:**

- Python (Jupyter/Colab)
- ChatGPT for code explanation/debugging

**Mini Project:**

- Basic data scraper & analyzer
- Python Quiz App

🎤 **Mock Interview:** Python + Logic + File Handling

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## ✅ Module 2: Data Handling with Pandas & Numpy (4 weeks)

**Duration:** Month 2

**Topics:**

- NumPy: arrays, indexing, broadcasting
- Pandas: DataFrames, Series, filtering, groupby
- Missing data handling
- Data wrangling (reshape, merge, pivot)

**Tools:**

- Pandas, NumPy, ChatGPT, Google Colab

**Mini Project:**

- IPL or Netflix dataset analysis using Pandas

🎤 **Mock Interview:** Data Manipulation + Real-world Datasets

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## ✅ Module 3: SQL & Excel for Analysts (3 weeks)

**Duration:** Month 3

**Topics:**

- SQL: SELECT, JOINs, GROUP BY, Subqueries, Window Functions
- Excel: formulas, pivot tables, charts, lookup, dashboards
- SQL for case studies

**Tools:**

- MySQL/PostgreSQL (pgAdmin), Excel, Formula Bot

**Mini Project:**

- Employee or Sales DB case study
- Excel Dashboard (e.g. Superstore or Ecommerce)

🎤 **Mock Interview:** SQL Joins + Case Study Discussion

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## ✅ Module 4: Data Visualization & Storytelling (3 weeks)

**Duration:** Month 4

**Topics:**

- Matplotlib, Seaborn, Plotly
- Creating interactive dashboards
- Power BI or Tableau (choose one)
- Storytelling with data

**Tools:**

- Power BI

**Mini Project:**

- Power BI Dashboard (e.g. COVID, Sales, HR analytics)

🎤 **Mock Interview:** Visualization Tools + Data Storytelling

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## ✅ Module 5: Statistics & Probability for Data Science (4 weeks)

**Duration:** Month 5

**Topics:**

- Descriptive statistics
- Probability distributions
- Sampling, CLT
- Hypothesis testing, t-tests, chi-square
- Correlation & Regression

**Tools:**

- Python (Scipy, Statsmodels), ChatGPT for formulas

**Mini Project:**

- A/B Testing Simulation
- Customer Churn Statistical Analysis

🎤 **Mock Interview:** Stats Scenarios + Probabilities + Tests

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## ✅ Module 6: Machine Learning – I (Supervised Learning) (4 weeks)

**Duration:** Month 6

**Topics:**

- ML pipeline
- Regression: Linear, Logistic
- Decision Tree, Random Forest, KNN
- Train-test split, model evaluation

**Tools:**

- Scikit-learn, Google Colab, ChatGPT, PyCaret (optional)

**Mini Project:**

- Loan Approval or House Price Prediction
- Predict Diabetes from health dataset

🎤 **Mock Interview:** Supervised Learning Models + Metrics

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## ✅ Module 7: Machine Learning – II (Unsupervised & Feature Engineering) (3 weeks)

**Duration:** Month 7

**Topics:**

- KMeans Clustering
- Dimensionality Reduction: PCA
- Feature selection, encoding, scaling
- Model tuning (GridSearchCV)

**Tools:**

- Scikit-learn, Seaborn, Colab

**Mini Project:**

- Customer Segmentation
- E-commerce Product Clustering

🎤 **Mock Interview:** Unsupervised Learning + Features

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## ✅ Module 8: Time Series + Intro to Deep Learning (3 weeks)

**Duration:** Month 8

**Topics:**

- Time Series components, lag, rolling stats
- Forecasting: ARIMA, Prophet
- Intro to Neural Networks
- Perceptron, basic Keras model

**Tools:**

- Statsmodels, Prophet, Keras, TensorFlow

**Mini Project:**

- Stock Price or Weather Forecasting

🎤 **Mock Interview:** Forecasting + Neural Net Basics

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## ✅ Module 8.5: Big Data with PySpark (2 weeks)

**Duration:** Between Month 8–9

**Topics:**

- What is Big Data?
- PySpark architecture & setup
- DataFrames, RDDs, Transformations, Actions
- PySpark SQL + basic MLlib
- Databricks (optional)

**Tools:**

- PySpark, Google Colab (Findspark), Databricks

**Mini Project:**

- Analyze NYC Taxi Dataset using PySpark

🎤 **Mock Interview:** PySpark Concepts + Big Data Thinking

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## ✅ Module 9: Natural Language Processing (NLP) (4 weeks)

**Duration:** Month 9

**Topics:**

- Text cleaning, tokenization
- TF-IDF, word embeddings
- Sentiment analysis
- Text classification (Naive Bayes, Logistic)

**Tools:**

- NLTK, SpaCy, Scikit-learn

**Mini Project:**

- Sentiment analysis on Twitter or Reviews
- Resume Classifier

🎤 **Mock Interview:** NLP Use Cases + Classification

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## ✅ Module 10: GenAI Foundations (4 weeks)

**Duration:** Month 10

**Topics:**

- What is GenAI & LLMs
- Prompt engineering
- Embeddings, Vector DBs (FAISS)
- OpenAI API, Anthropic, Gemini

**Tools:**

- OpenAI, ChatGPT, Claude, Gemini, Replit, Colab

**Mini Project:**

- Build a chatbot using OpenAI
- Auto-summarizer using ChatGPT API

🎤 **Mock Interview:** Prompting, GenAI APIs, RAG Concept

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## ✅ Module 11: GenAI App Building (4 weeks)

**Duration:** Month 11

**Topics:**

- LangChain basics
- LlamaIndex, RAG pipeline
- Build workflows with Flowise
- Deploy with Streamlit/Gradio

**Tools:**

- LangChain, Streamlit, Hugging Face, Vercel

**Mini Project:**

- Resume Q&A bot using LangChain
- Voice-to-Summary App using Whisper + GPT

🎤 **Mock Interview:** LLM App Architecture + RAG Debugging

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## ✅ Module 12: GenAI Tools Lab & AI Playground

Duration: Month 11

🎯 Goal: Hands-on practice with advanced AI tools using Python – build, integrate, and deploy mini AI solutions.

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### 🔷 Key Focus Areas

- **AI Tools with Python:**
  - 🔷 OpenAI API (ChatGPT, Whisper, DALL·E)
  - 🔷 Hugging Face (Transformers, CLI)
  - 🔷 LangChain (for chatbots & automation)

- 🔷 PandasAI (dataframe + GPT)
- 🔷 AutoGPT / BabyAGI (autonomous agents)
- 🔷 Replicate (ML models via API)
- 🔷 LlamaIndex (custom document Q&A)
- 🔷 Weights & Biases (experiment tracking)

- **Mini Project Lab:**
  - Resume screener, report generator, image-to-text, etc.
- **Cloud Tools:**
  - Google Colab, Hugging Face Spaces, Streamlit Cloud
- **Capstone Activity:**
  - Build & demo your own GenAI tool using 2+ APIs
- **Mock Interview (Tool-Based)**
  - Questions + project walkthrough focused on AI tools

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## ✅ Module 13: Portfolio + Mock Interviews + Job Prep (Final Month)

**Duration:** Month 12

**Topics:**

- Resume building (AI optimized)
- Mock interviews (DA, DS, GenAI)
- LinkedIn & GitHub projects
- Capstone project (group/solo)
- Freelancing & internships guide

**Capstone Projects (Pick 1-2):**

- Data Analytics Dashboard (end-to-end BI project)
- ML Web App (e.g., Predict & display output with Streamlit)
- LLM Chatbot with LangChain + Vector DB
- YouTube Transcript Analyzer using Whisper + GPT

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